QIANG GUO

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EDUCATION

The Hong Kong University of Science and Technology (HKUST) – MSc, Electronic Engineering **Sep 2018 – Nov 2019**
Nanjing University of Aeronautics and Astronautics (NUAA) – BEng, Aircraft Power Engineering **Sep 2014 – Jun 2018**
National Tsing Hua University (NTHU), Taiwan – Exchange, Mechanical Engineering **Sep 2016 – Jan 2017**

RESEARCH INTERESTS

AI for credit risk and financial decision-making; causal inference and experimentation; machine learning under distribution shift and cold start.

PROFILE

~7 years of applied modeling, experimentation and decision-making in consumer and SME credit at Ant Group, ByteDance and Du Xiaoman. Seeks doctoral training to give this practice rigorous methodological foundations.

RESEARCH-RELEVANT EXPERIENCE

Ant Group – Ant Consumer Finance (Huabei Installments), Beijing **Dec 2025 – Present**
Senior Expert, Risk Strategy · Credit line and pricing strategy for a consumer installment portfolio

- **Experimental design & causal inference (pricing):** Designed an unbiased, statistically powered, platform-wide price-discount experiment to estimate segment-level discount ROI (lifecycle × risk × creditworthiness) and locate its inflection point, with AI-automated data collection and monitoring. Paired it with risk-based price increases for price-insensitive users that offset margin loss, managed against finance targets, and defined requirements for price-response models.
- **Cold start: data enrichment & identification:** Built the credit strategy for a new vertical, lithium-battery financing for heavy-truck drivers. This small segment had not been managed granularly and lacked business-operation signals (identity, business cash flow, operating capability). Enriched the data by fusing manufacturer-provided and internal risk data, built self-employed/SME operator identification with AI-assisted feature mining, and refit models. The result raised approval rate and full-line coverage at stable risk.
- **Network-based modeling:** Used AI to identify, generate and match supply-demand networks that test whether transaction needs are plausible in higher-risk offline merchant settings. Compressed them into deployable decision rules backed by optimal model refits; the method is now in production and selectively raises full-line coverage.

ByteDance, Beijing **Dec 2022 – Aug 2025**
SME Credit Risk Strategy Expert · Led SME pre-loan credit-line strategy and merchant in-loan line-increase; SME alternative-data point of contact

Constrained Pre-Loan Line Optimization under Creditworthiness × Risk (led a 2-person virtual team; Dec 2022 – Dec 2023, 9 iterations): raise approval rate and credit line under a fixed risk constraint

- **Methods:** Designed a creditworthiness × risk layered framework with dual-layer lines (fixed + temporary). Closed loop: launch → cell-by-cell review → attribution → framework adjust, with routine A/B expand/rollback. Large-ticket admission via multi-model cross-check plus bureau/liability rules; upgraded static matrix to incremental line-setting + income-multiple caps. Onboarded external data (insurance, financial statements) to upgrade quality tiers.
- **Results:** Approval rate +50% (relative); credit line 2x; avg ticket ~+75%; monthly originations +30%; risk ~−18% to −20%; balance 15x in one year. Large-ticket avg nearly 2x with risk structure −>60%; top-tier share nearly 2x; external-data ROI >20x.

In-Loan Line-Increase via GMV × Creditworthiness × Risk (led a 2-person virtual team; Apr 2024 – May 2025, ~10 iterations): CNY 70B+ portfolio, 4M+ customers; grow first-draw volume and balances while controlling risk on CNY 500K+ exposures

- **Methods:** Designed a GMV × creditworthiness × risk uplift method with multi-dimensional caps. Multi-source GMV fusion (external e-commerce, acquiring, statements, annual reports, tax invoices, operating-income model) with confidence coefficients and time decay; line-setting for externally matched populations. Built a seasonal/promo line-increase system from 0→1. Large-ticket admission from delinquency sample review; merchant creditworthiness as scale × stability × profile; A/B on fixed vs temporary line mix.
- **Results:** Segment risk −30%; outstanding balance 2x in one year; average line +CNY 50K; CNY 500K+ exposure risk at 0.25x the portfolio level. Uplift-cohort avg line +30–40% with small risk increase; thick-data cohort ~+60% in half a year; large-ticket uplift risk ~−60%; admission +~4pp approval / −~10% risk; stratified retention +~6pp; treatment conversion +~10% (relative).

Alternative-Data Evaluation for SME Credit: ROI evaluation and onboarding of 8 external data types (owner–enterprise linkage, business registry, annual reports, merchant acquiring, e-commerce, ad spend, tax invoices, customs) for SME profiling, creditworthiness and GMV estimation; enterprise (business-registry) and tax-invoice data had the largest impact; selected sources (insurance, financial statements) ROI >20x

- **Methods:** Designed the test plans and wrote evaluation reports. Ran the pipeline from research and testing through procurement, data warehousing and production, working with clients, legal, data product and vendors, and sourced internal SME scenario data across departments.

Du Xiaoman Financial, Beijing **Oct 2021 – Dec 2022**

Risk Strategy Expert (internal transfer & promotion)

Feature Construction for SME Credit Lines: differentiate small-business owners within non-scenario consumer lending at stable risk

- **Methods:** Built SME scale and stability features from business-registry data (industry, location, registered capital, shell-company flags) and offline media activity, and validated them against profiles and macro data. Raised lines for stable, high-value SMEs and cut exposure to unstable or high-risk ones.
- **Results:** Initial credit line +10%; first-loan T30 average balance +5%; relative risk −30%.

Response Modeling & a Field Experiment on Undrawn Credit **(1-person workstream)**: offer-allocation efficiency within 30 days of approval; response and risk after approvals expire

- **Methods:** Used high-value, response and price-sensitivity models to reallocate line and price offers, and an offline low-risk recall model to re-approve expired customers. Ran a validity-extension experiment that moved risk screening to the transaction step to address data staleness and adverse selection.
- **Results:** Utilization +5% and T30 balance retention +1% within validity; for expired customers, risk −5% and utilization +10%.

Lifecycle-Based Account Management (post-payoff re-engagement): fed credit-enhancement signals and response/sensitivity models into line-increase and price-cut decisions at the payoff point. Repeat borrowing +6% (relative)

Du Xiaoman Financial, Beijing **Jun 2019 – Oct 2021**

Senior Algorithm Engineer · Fraud and liveness programs prevented CNY 100M+-level potential losses

Telecom-Fraud Detection **(built 0→1 in 2019; grew to 3-person team in 2020)**: end-to-end detection system; set KPIs for strategy, feature mining and modeling, with reviews alongside credit review, complaints and security

- **Methods:** Developed 3 generations of detection models (basic scam patterns, large-loan differentiation, real-time features). Mined 20+ highly predictive features from device and app-list data. Used risk tiering and finer splits of sparse large-loan segments. Built a T+1 case-monitoring platform, plus real-time feature services that brought features online 2 weeks sooner.
- **Results:** Fraud recall 90%+; fraud incidence 0.01%; +10% recall from new features; manual review volume ~67% lower with +30% recall; 2 large organized medical-aesthetics fraud rings stopped; Company Innovation Award (two consecutive quarters).

Liveness Detection **(cross-functional)**: low fraud catch rate, fraud losses, imbalanced manual-review capacity and weak model efficiency

- **Methods:** Broke down anti-fraud KPIs into liveness eligibility and evaluation metrics. Early on, traded manual-review capacity to ease attack pressure; later, built fraud features from liveness and OCR images and iterated toward full-scenario coverage.
- **Results:** Attacks from single digits per month to zero; impostor-detection precision 2x; manual review −85% at 99% recall.

SKILLS

Technical: SQL, Python, Shell, Java, Hadoop, Spark (proficient: SQL, Python, Shell, Hadoop; familiar: Java, Spark); experiment design, ML modeling, feature engineering, network-based features

Other: PMP · English (professional), Mandarin (native), Spanish (A1)